SERHAT TADIK

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EDUCATION

Georgia Institute of Technology — Aug 2022 - Present
PhD in Applied ML for Wireless Communications — GPA: 4/4
School of Electrical & Computer Engineering

Bogazici University — Sep 2016 - Feb 2022
Undergraduate Degree — Overall GPA: 3.79/4
Department of Electrical & Electronics Engineering, Double Major in Physics

Washington University in St. Louis — Jan - June 2020
Student Exchange Program — *SPA: 3.94/4*

TECHNICAL STRENGTHS

Computer Languages	Python, C, C++, MATLAB, SQL
Frameworks	PyTorch, PyG, Tensorflow, Scikit-learn, FastAPI, pytest, Sagemaker, Lambda
Tools	Bash, git, Docker, AWS, SQL Databases, GIS, Claude Code, OpenAI Codex

PUBLICATIONS & REVIEW

Journal Publications

- Augmented RF Propagation Modeling
  S. Tadik *et al.*
- Digital Spectrum Twins for Enhanced Spectrum Sharing and Other Radio Applications
  S. Tadik *et al.*
- Reflection of Modulated Radio: System-level Design, Analysis, and Performance Evaluation for Ambient Scatter Communication Systems
  M. A. Varner, ***S. Tadik****, ...*

Conference Publications

- OpenGERT: Open Source Automated Geometry Extraction with Geometric and Electromagnetic Sensitivity Analyses for Ray-Tracing Propagation Models
  S. Tadik *et al.*
- How Critical is Site-Specific RAN Optimization? 5G Open-RAN Uplink Air Interface Performance Test and Optimization from Macro-Cell CIR Data
  J. Corgan, N. Nair, R. Bhattacharjea, W. Liu, ***S. Tadik****, T. Tsou, T. J. O'Shea*
- Georeferenced Spectrum Occupancy Analysis Using Spatially Very Sparse Spectrum Monitoring Data
  S. Tadik *et al.*
- POWDER-RDZ: Prototyping a Radio Dynamic Zone using the POWDER platform
  D. Johnson, D. Maas, ***S. Tadik****, ...*
- G-HCF: Product Recommendation by GNN-Based Hybrid Collaborative Filtering
  S. Tadik *et al.*

Reviewer Activities

- IEEE Transactions on Cognitive Communications and Networking
- IEEE Transactions on Intelligent Vehicles
- IEEE Journal of Radio Frequency Identification

EXPERIENCE

Polaris Wireless, USA Aug 2024 - Present
Received PhD Funding Twice, DSP & ML Intern

- Leading efforts to modernize and optimize a cellular network-based location estimation engine, developing advanced feature extraction pipelines for processing time-series network measurement data, timing advance information, and propagation maps. Achieved a 6.1% reduction in Euclidean error compared to legacy systems.
- Designed and implemented advanced machine learning algorithms for location-based monitoring and detection systems. Developed novel training methodologies that demonstrated superior performance in challenging data scenarios with limited samples, achieving up to 14.3% improved classification accuracy compared to traditional approaches. Created an innovative confidence scoring system using learned feature representations that, when combined with optimized thresholds, enabled the model to achieve 100% classification accuracy on key performance metrics. The designed model architecture and implementation are under consideration within the company for patent filing or trade-secret status.

Deepsig, USA May - Aug 2024
DSP & ML Intern

- Implemented an optimal 2D filter for LTE channel estimation. Developed algorithms for Doppler and delay spread estimations.
- Worked on data preparation and novel loss function design for a specific type of generative channel modeling.
- Worked on time and position interpolation of real-world LTE channel measurements.
- Computed channel statistics and time-of-flight for the measurements and generated a set of ground truth channels.
- Co-authored a paper named *How Critical is Site-Specific RAN Optimization? 5G Open-RAN Uplink Air Interface Performance Test and Optimization from Macro-Cell CIR Data.*

Cognosos Inc., USA Jan - May 2024
Research and Development Intern

- Working on indoor zone classification using ANN, CNN, and pre-ML techniques such as XGBoost to analyze sources of misclassification. Developed a hierarchical model combining an ANN with a One-vs-One classifier, improving accuracy by 1%.
- Implemented various voting algorithms for more stable and accurate predictions, improving accuracy by at least 0.5%.
- Demonstrated that combining samples at training and inference stages improves accuracy by up to 1.6% by eliminating noise and nulls in signal strength distributions.
- Developed a radio-mapping diagnostics tool to detect outliers, increasing the reliability of data used for training. Led data collection campaigns to analyze tool effectiveness and determine optimal transmitter density for maximum accuracy.

The Propagation Group @ Georgia Institute of Technology Aug 2022 - Present
Graduate Research Assistant

- Developed methods for quantifying and visualizing spectrum occupancy from spectrum monitoring data, introducing new spectrum occupancy visualizations for future allocation and interference analysis.

- Contributed to the implementation of an ambient scatter communications system, employing a complex-valued neural network demodulation technique that surpasses traditional methods.
- Enhanced electromagnetic wave propagation models by applying regularized regression-based error correction, significantly reducing error variance by 58% to 87%.
- Designed digital spectrum twins for radio spectrum usage tracking and prediction, utilizing GIS, OSM, Python, and MATLAB. The models combine received signal strength, variance, duty cycle, and confidence values for various radio users in a region of interest.

Caretta Software, Turkey Oct 2021 - Sep 2022
Machine Learning Engineer

- Implemented feature design & engineering, customer segmentation, and churn prediction analysis. Applied clustering algorithms like KMeans and DBSCAN, gaining insights into customer behavior.
- Utilized PyTorch to train an ANN model for churn prediction, recovering 26% of potentially churned customers with targeted promotions.
- Led a project on customer product recommendation using graph neural networks, achieving 90% accuracy in transaction prediction with a GAT model. Work accepted for publication in IEEE Signal Processing and Communications Applications Conference.

Bogazici University, Turkey Sept 2018 - Jan 2020
Undergraduate Research Assistant

- Completed an undergraduate thesis on "A Comparison Between GNN Architectures and Implementation on Brain Connectomes", demonstrating the superiority of GIN over GCN in graph classification tasks.

PROJECTS

Data Analytics & Visualization Platform 2024 - Present
Independent Development Project

- Designed and developed a comprehensive data analytics and visualization platform leveraging LLM APIs and local language models to provide intelligent insights for enterprises to analyze their proprietary databases and internal data assets.
- Built scalable backend microservices architecture using Python and FastAPI, implementing authentication, session management, LLM, search, and export services with inter-service communication.
- Developed responsive frontend interface using JavaScript, HTML, and CSS for intuitive data visualization and user interaction.
- Implemented robust infrastructure including Kong API Gateway for service orchestration, Redis for caching and session management, and PostgreSQL for data persistence.
- Demonstrated end-to-end product development skills from system architecture design to deployment-ready implementation.

RELEVANT COURSES

Wireless Networks	Personal & Mobile Communications
Probabilistic Graphical Models in ML	Information Theory
Generative and Geometric Deep Learning	Random Processes and Kalman Filtering

ACHIEVEMENTS

- Have received PhD funding from Polaris Wireless for 2 consecutive years during my PhD program.
- Provided a brown-bag talk at Polaris Wireless regarding the efforts to modernize and improve the geofence violation technology led by me.

- Was nominated for the employee of the month as an intern at Cognosos, Inc.
- Received a best-paper award with the paper that I co-authored (POWDER-RDZ*: Prototyping a Radio Dynamic Zone using the* POWDER *platform*) in IEEE DySPAN 2024.
- Graduated with high honors from the Electrical Engineering and Physics Departments at Boğaziçi University.
- Ranked 36^{th} among 2,000,000 candidates in the university entrance exam LYS, in 2016, in Turkey